

Red Team Engagement Report - 8. Comprehensive Reporting Lab

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This red team engagement simulated realistic adversary techniques to evaluate organizational resilience. The exercises included phishing, fileless execution with PowerShell, and credential harvesting through WMI and Mimikatz. Each phase demonstrated how attackers can exploit legitimate Windows tools to bypass antivirus and gain elevated access. While the attacks were conducted in a controlled lab environment, they highlight real risks that organizations face daily. Key findings emphasize the importance of enforcing multi-factor authentication, restricting PowerShell usage, and monitoring WMI activity. Implementing these measures will significantly reduce exposure to credential theft, privilege escalation, and domain compromise.

Findings

- Phishing (T1566) → simulated email attack
- Fileless Execution (PowerShell) → payload executed in memory
- Credential Harvest (WMI/Mimikatz) → credentials dumped

Recommendations

- Enforce multi-factor authentication (MFA)
- Restrict or monitor PowerShell usage
- Disable or monitor remote WMI execution
- Detect suspicious LSASS access

Finding ID	TTP	CVSS Score	Remediation
FID001	Phishing (T1566)	7.5	MFA enforcement
FID002	Fileless Execution	8.0	Restrict PowerShell use
FID003	Credential Harvest	9.0	Monitor WMI, LSASS

Findings

During the engagement, several attack phases were simulated:

- Phishing (T1566): A phishing email was used to demonstrate how attackers gain initial access.
- Fileless Execution: PowerShell was leveraged to run payloads directly in memory, bypassing antivirus detection.
- Credential Harvest: WMI was used to launch Mimikatz, exposing privileged account credentials.

Attack Path Diagram

